

MARION COUNTY AP, SC, USA

WMO: 720608

Lat: 34.181N

Lon: 79.335W

Elev: 28

StdP: 100.99

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00200

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.0	(c) -2.3	(d) -13.9	(e) 1.1	(f) -0.8	(g) -11.1	(h) 1.4	(i) 1.2	(j) 10.5	(k) 15.7	(l) 8.8	(m) 12.0	(n) 1.6	(o) 20	(p) 0.375

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	35.0	24.2	33.0	24.0	32.4	23.9	26.1	30.5	25.6	30.2	25.2	29.8	3.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.3	27.1	24.8	19.9	27.1	23.8	18.7	26.8	81.3	29.8	78.5	30.8	76.6	29.2	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.3	7.1	5.8	-7.9	36.1	1.9	1.6	-9.2	37.2	-10.3	38.2	-11.4	39.0	-12.8	40.1
			-9.1	26.8	2.0	0.6	-10.5	27.3	-11.7	27.6	-12.8	27.9	-14.3	28.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	7.3	9.3	12.3	17.4	21.9	25.6	27.1	26.3	23.8	18.0	11.7	9.9
	DBStd	8.04	5.60	5.16	5.18	4.07	3.38	2.51	1.91	1.76	2.67	4.09	4.46	4.82
	HDD10.0	323	120	71	36	2	0	0	0	0	0	1	30	63
	HDD18.3	1395	343	256	197	64	11	0	0	0	1	55	203	265
	CDD10.0	3096	36	51	107	224	368	468	529	506	415	250	81	61
	CDD18.3	1127	1	3	9	36	121	218	271	247	166	46	4	4
	CDH23.3	9980	4	20	91	338	978	2052	2667	2179	1277	339	20	13
	CDH26.7	3993	0	1	11	71	325	896	1232	914	472	72	1	0
Wind	WSAvg	2.3	2.5	2.8	2.9	3.0	2.5	2.1	2.2	1.8	1.9	2.1	2.0	2.2
Precipitation	PrecAvg	1220	93	82	92	74	91	122	138	158	128	90	79	79
	PrecMax	1489	183	219	179	137	168	219	241	304	359	194	157	172
	PrecMin	849	30	33	20	11	28	28	58	86	24	2	5	26
	PrecStd	201	50	41	38	33	30	46	48	61	89	54	42	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.7	26.0	27.8	30.2	35.0	36.4	37.0	35.0	33.0	31.2	26.0	25.1
		MCWB	18.1	18.1	17.7	20.1	22.6	24.0	25.0	24.8	22.9	23.4	19.3	20.2
	2%	DB	21.2	22.8	26.1	28.1	32.0	34.1	35.1	32.9	32.3	28.0	22.9	22.6
		MCWB	16.8	17.1	17.2	18.8	21.8	23.9	24.6	24.5	23.0	21.1	17.6	18.7
	5%	DB	18.0	21.1	23.0	27.3	30.1	32.6	33.0	32.4	31.2	27.1	21.4	20.9
		MCWB	14.7	15.6	15.6	18.7	21.6	23.4	24.5	24.3	23.0	20.4	17.0	18.4
	10%	DB	17.0	17.9	21.3	25.2	27.9	31.4	32.4	31.4	29.9	25.1	20.0	18.0
		MCWB	13.8	14.0	15.2	17.6	20.9	23.1	24.5	24.1	22.9	19.2	16.6	16.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.1	19.4	19.9	22.8	24.3	26.0	27.2	26.8	25.7	24.1	21.2	21.3
		MCDB	21.6	23.1	23.5	26.2	30.1	32.0	32.1	30.8	28.6	28.6	23.0	23.3
	2%	WB	17.9	18.4	18.9	21.5	23.3	25.3	26.3	26.1	24.9	23.3	19.9	20.1
		MCDB	19.8	22.4	23.2	25.2	28.4	31.4	31.0	29.9	28.8	26.8	21.4	22.0
	5%	WB	16.5	17.0	17.9	20.6	22.7	24.7	25.7	25.6	24.2	22.3	18.5	18.5
		MCDB	18.2	19.8	21.3	24.1	27.3	30.3	30.6	29.6	28.5	25.0	21.2	20.1
	10%	WB	14.0	15.0	16.8	19.5	22.1	24.0	25.2	25.1	23.7	21.1	17.3	16.8
		MCDB	16.2	17.7	20.0	23.4	26.5	29.2	30.1	29.2	27.7	23.3	18.8	18.2
Mean Daily Temperature Range	MDBR		11.0	11.6	12.3	12.7	11.2	10.8	10.6	9.8	10.1	11.3	12.1	10.5
	5% DB	MCDBR	13.7	13.8	14.4	14.4	13.0	12.1	11.6	11.1	11.8	12.4	13.1	11.9
		MCWBR	9.3	8.4	7.1	6.3	4.6	3.9	3.7	3.7	4.1	5.9	8.1	7.9
	5% WB	MCDBR	11.7	12.8	11.8	11.0	10.6	10.9	10.5	9.5	9.6	10.0	10.5	10.1
		MCWBR	9.3	8.9	6.8	5.9	4.3	4.1	3.9	3.7	3.8	6.0	7.8	7.9
Clear-Sky Solar Irradiance	taub		0.319	0.322	0.362	0.401	0.465	0.489	0.538	0.509	0.430	0.392	0.343	0.322
	taud		2.506	2.494	2.407	2.285	2.162	2.138	2.022	2.092	2.307	2.380	2.475	2.530
	Ebn at Noon		893	929	912	886	830	806	765	780	832	836	855	868
	Edh at Noon		84	94	112	132	151	155	173	158	122	105	86	78
All-Sky Solar Radiation	RadAvg		2.74	3.38	4.50	5.67	6.16	6.40	6.11	5.51	4.71	3.92	3.02	2.41
	RadStd		0.15	0.35	0.37	0.38	0.48	0.43	0.38	0.27	0.39	0.46	0.24	0.20

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (63 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+170	+122

Nomenclature: See separate page